

SWAPNANIL DEY

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PROFESSIONAL SUMMARY

Generative AI Engineer with nearly 3 years of experience building production-grade LLM applications, Retrieval-Augmented Generation (RAG) systems, AI assistants, and scalable backend services. Hands-on experience with Azure OpenAI, LangChain, Python, Qdrant, FastAPI, Langfuse, and MLflow across the full GenAI lifecycle including data ingestion, embeddings, retrieval, prompt engineering, evaluation, observability, and deployment. Developed enterprise AI solutions integrating SharePoint, Confluence, ServiceNow, and Bitbucket to enable intelligent knowledge discovery and self-service support. Experienced in mentoring team members on GenAI concepts and delivering business-focused AI solutions in large-scale enterprise environments.

EDUCATION

Bachelor of Technology in Computer Science and Engineering, MAKAUT

July 2019 - July 2023

Techno Main Salt-Lake

Silver Medalist

Grade - CGPA 9.28

SKILLS

GenAI & LLMs: Azure OpenAI, LangChain, Prompt Engineering, Chain-of-Thought (CoT) Reasoning, Retrieval-Augmented Generation (RAG), Embeddings, LLM Evaluation, AI Assistants, Semantic Search

ML & NLP: Machine Learning, NLP, Azure Computer Vision, OCR

Backend & APIs: Python, Flask, FastAPI, REST API Development

Data & Infrastructure: Qdrant (Vector DB), PostgreSQL, AWS, Langfuse, MLflow, Vector Search

Enterprise Tools: JIRA, ServiceNow, Confluence, SharePoint, Bitbucket

Soft Skills: Problem-Solving, Communication, Collaboration, Adaptability, Critical Thinking, Leadership

EXPERIENCE

Systems Engineer (GenAI Developer)

August 2023 - Present

TCS

Kolkata, India

Project: Technology Service Copilot

Tech Stack: Python, Azure OpenAI, LangChain, Langfuse, MLflow, Qdrant, AWS, FastAPI, PostgreSQL

- Built a Retrieval-Augmented Generation (RAG) pipeline supporting 10000+ internal documents integrating Service Catalogs, ServiceNow, Confluence, Bitbucket and SharePoint to power a GenAI virtual assistant for enterprise tech support reducing support resolution time by 40%.
- Designed and implemented a real-time, SSE (Server-Sent Events)-based step-streaming pipeline that surfaces backend processing stages (query analysis, hypothetical question generation, chunk retrieval, answer generation) live to the frontend, improving transparency and user trust during response generation.
- Implemented end-to-end embedding pipelines into Qdrant VectorDB with automated, scheduled ingestion and audit logging, reducing manual data-prep effort by 40%.
- Built automated AWS S3 ingestion connectors (Bitbucket markdown extractor, Sharepoint and Asset Management PDF fetcher), cutting manual ingestion effort by 70%.
- Contributed (60%) to a feedback and FAQ analytics module enabling content owners to track user feedback trends and frequently asked questions in real time.
- Integrated observability and evaluation frameworks using Langfuse and MLflow for monitoring LLM performance, traceability, and response quality analysis.
- Collaborated with cross-functional teams to deploy scalable GenAI solutions supporting real-time enterprise knowledge discovery and self-service assistance.

Project: Knowledge Assistant

Tech Stack: Azure OpenAI, Azure Computer Vision, LangChain, Python, Flask, Cosmos DB

- Developed a document/URL ingestion pipeline with OCR (Azure Computer Vision) and LangChain-based chunking and embedding for QnA and summarization use cases.
- Built feedback collection and source-attribution features to improve response transparency and trust.
- Applied prompt engineering to generate structured product profiles from business documents.

Project: Model-Agnostic Explainable AI & Bias Detection System

Tech Stack: Python, Flask

- Built REST APIs integrating local, global, fairness/bias, and counterfactual explanation modules with GenAI-based natural language narration.
- Implemented local explanation support for image classification models (.keras, .pkl).

Intern
TCS

Feb 2023 - May 2023
Kolkata, India

Project: Dashboard for Job Run Status

Tech Stack: Python, ReactJS, Flask

- Built CRUD operations on Excel-based job run data and developed interactive dashboards (pie charts, tables) for internal monitoring.

CERTIFICATIONS

Microsoft Certified: Fabric Analytics Engineer Associate	May 2024
Microsoft Certified: Azure Data Engineer Associate	October 2023
Machine Learning Specialization	February 2021

AWARDS

Innovation Pride Award For contributing an innovative idea recognized during the TCS Innovista 2026 innovation program	May 2026
Applause for Team Award For delivery excellence and outstanding customer satisfaction through consistent team dedication in GenAI solutions.	June 2025
Star Team Award for delivering innovative GenAI-driven solutions for our customer.	Jan 2025
Contextual Master Award for leveraging contextual knowledge to create tools that enhance operational efficiency with AI powered Knowledge Management System.	Dec 2024
Star Team Award for delivering innovative GenAI-driven solutions for our customer.	July 2024

LANGUAGES

- English
- Hindi
- Bengali